

Solving Word Problems Involving Quadratic Functions and Equations

Jonathan R. Bacolod

Quadratic equations may be solved by **factoring**,
completing the square, the **quadratic**
formula, or by using the **graph** of the function.

Methods of Solving

- ▶ **Factoring** — write $ax^2 + bx + c = 0$ as a product of factors.
- ▶ **Completing the square** — rewrite the equation in the form $(x - h)^2 = k$.

- ▶ **Quadratic formula** —

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

- ▶ **Graphing** — find where the graph crosses the x -axis.

Zeros or Roots

The **zeros** or **roots** of a quadratic function are the values of x where $f(x) = 0$.

$$f(x) = x^2 - 5x + 6$$

Setting $f(x) = 0$ and solving gives $x = 2$ and $x = 3$. These are the zeros of the function.

Modeling Real-Life Situations

Many **real-life situations** can be modeled using quadratic equations and functions, such as:

- ▶ **Area** problems (gardens, frames, floors)
- ▶ **Motion** problems (falling or thrown objects)
- ▶ **Geometry** problems (dimensions of shapes)
- ▶ **Business** problems (profit, revenue, cost)

Interpreting Solutions

Always **interpret the answer** in the context of the given problem.

Reject **impossible values** when necessary — for example, a negative length or a negative time does not make sense in real life.

Examples

Example 1: Area

The area of a square garden is 144 m^2 . Find the length of one side.

$$s^2 = 144$$

$$s = \sqrt{144} = 12$$

The side length is 12 m.

Example 2: Dimensions

A rectangle's length is 3 m longer than its width. Its area is 70 m^2 . Find its dimensions.

$$w(w + 3) = 70$$

$$w^2 + 3w - 70 = 0$$

$$(w - 7)(w + 10) = 0$$

Since width cannot be negative, $w = 7$. The width is 7 m and the length is 10 m.

Example 3: Quadratic Formula

Solve using the quadratic formula:

$$2x^2 + 3x - 5 = 0$$

$$x = \frac{-3 \pm \sqrt{3^2 - 4(2)(-5)}}{2(2)} = \frac{-3 \pm \sqrt{49}}{4}$$

$$x = 1 \quad \text{or} \quad x = -\frac{5}{2}$$

Practice Exercises

Practice 1

The area of a square is 169 cm^2 . Find the length of one side by writing and solving a quadratic equation.

Practice 2

A rectangle has a length that is 5 m longer than its width. If its area is 84 m^2 , find its dimensions.

Practice 3

A ball is thrown upward and its height after t seconds is

$$h(t) = -5t^2 + 20t + 2.$$

After how many seconds will the ball hit the ground?

Practice 4

The product of two consecutive integers is 156.
Find the integers.

Practice 5

A garden has a length that is 4 m more than its width. Its area is 96 m^2 . Find its dimensions.

Practice 6

Solve using the quadratic formula:

$$3x^2 - 7x - 6 = 0.$$

Practice 7

The profit of a small business is modeled by

$$P(x) = -2x^2 + 40x + 120,$$

where x is the number of items sold (in tens).
Find the maximum profit.

Practice 8

A projectile follows the path

$$h(t) = -4.9t^2 + 29.4t.$$

Determine when the projectile returns to the ground.

Activities

Activity 1

A rectangle has a length that is 7 m longer than its width. If its area is 144 m^2 , find its dimensions.

Activity 2

The product of two consecutive integers is 210.
Find the integers.

Activity 3

Solve using the quadratic formula:

$$2x^2 - 9x - 5 = 0.$$

Activity 4

A rectangular picture frame has an outer dimension of 20 cm by 28 cm. A uniform border of width x cm leaves a picture area of 288 cm^2 . Find the width of the border.

Activity 5

The sum of two numbers is 20 and their product is 84. Find the two numbers.

Thank you!